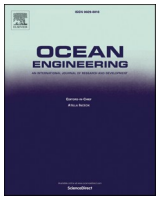
Ocean Engineering 322 (2025) 120400



Contents lists available at ScienceDirect

Ocean Engineering

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Research paper



Navigating narrow waterways: A quantitative study on timing of stand-on ship actions for cooperative collision avoidance for MASS

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ARTICLE INFO

Keywords:
Ship collision
Collision avoidance
CFD
CDC
COLREGS
MASS

ABSTRACT

Article 17 of the COLREGS is ambiguous as to when stand-on ships are required to take action to avoid colliding. This study will address this ambiguity by quantification of factors in a manner analogous to road traffic laws. Given the complex and variable nature of maritime situations, along with the diverse maneuvering capabilities of different ships, this task is inherently challenging. To tackle this issue, this study employs the KCS as a model ship to propose timing of collision avoidance action when the KCS is a stand-on ship. By utilizing computational fluid dynamics (CFD) to obtain the KCS's trajectory, we will seek to provide clear guidelines on when a stand-on ship should act for collision avoidance action. The resulting numerical values will be valuable not only for maritime autonomous surface ship (MASS) but also for manned ships, thereby contributing to improved navigation safety and more effective collision prevention strategies at sea.

1. Introduction

As we enter the fourth industrial era with the development of artificial intelligence (AI) technology, research on maritime autonomous surface ships (MASS) is active (Lee and Park, 2024). For the realization of MASS, collision avoidance research is one of the most important factors among many research themes (Gao and Shi, 2020; Chun et al., 2021; Lee and Park, 2022). Therefore, many researchers are conducting collision avoidance research using AI (Namgung, 2022; Lee and Park, 2024). Moreover, based on many AI techniques, collision avoidance research considering COLREGS is also being actively conducted (Lee and Park, 2022, 2024). This is necessary because all ships sailing at sea must perform collision avoidance actions following COLREGS (Tam and Bucknall, 2010; Hasegawa et al., 2012; Porathe, 2019; Huang et al., 2020; Lee and Park, 2024).

However, there are still many limitations to the research that has been presented, as many of these studies have not addressed the core issues of COLREGS (Lee et al., 2021). COLREGS is a qualitative regulation (Zhou et al., 2020; Wrobel et al., 2022). There are no specific numerical values for collision avoidance action (Rule 7). And Rule 17 simply states that stand-on ships should cooperate in collision avoidance action when two ships are close and collision cannot be avoided by the give-way ship's action alone. Very little research has been done to provide specific numerical values in this regard (Lee and Furukawa, 2022). Du et al. (2020a, 2020b) and Wang et al. (2021) studied for stand-on ship. However, their studies were based on the ship domain, which makes it difficult to consider them as an objective criteria for collision avoidance cooperation action of stand-on ships.

It is unfeasible to quantitatively regulate every situation that can occur at sea and present criteria values that considers all situations. However, criteria values for each rule of COLREGS should be provided. This is necessary for allowing seafarers to make judgments based on standardized criteria values in ambiguous situations they experience for the first time.

As the first step towards providing a solution, Lee and Furukawa (2022) studied the timing of collision avoidance actions of stand-on ships using KVLCC2. As mentioned earlier, COLREGS Rule 17 specifies that a stand-on ship should take avoidance cooperative action when a collision cannot be avoided solely by the actions of the give-way ship. The presented study sought to specify the action timing of the stand-on ship by developing a collision assessment (CA) model named computed distance at collision (CDC). This is a CA model that demonstrates high accuracy in situations where two ships are very close and collision avoidance action is dire. Therefore, ship collision simulations were conducted to identify situations where collisions between ships are inevitable and to provide an objective and quantifiable timing of

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https://doi.org/10.1016/j.oceaneng.2025.120400
Received 18 August 2024; Received in revised form 27 November 2024; Accepted 13 January 2025
Available online 27 January 2025
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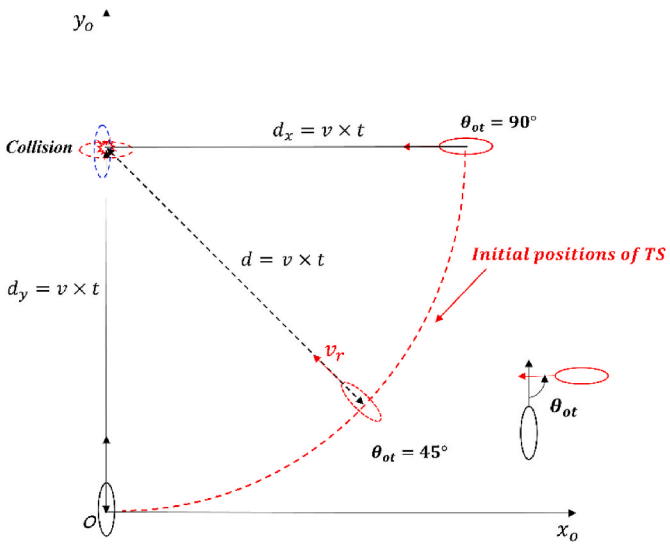


Fig. 1. Configure a scenario.

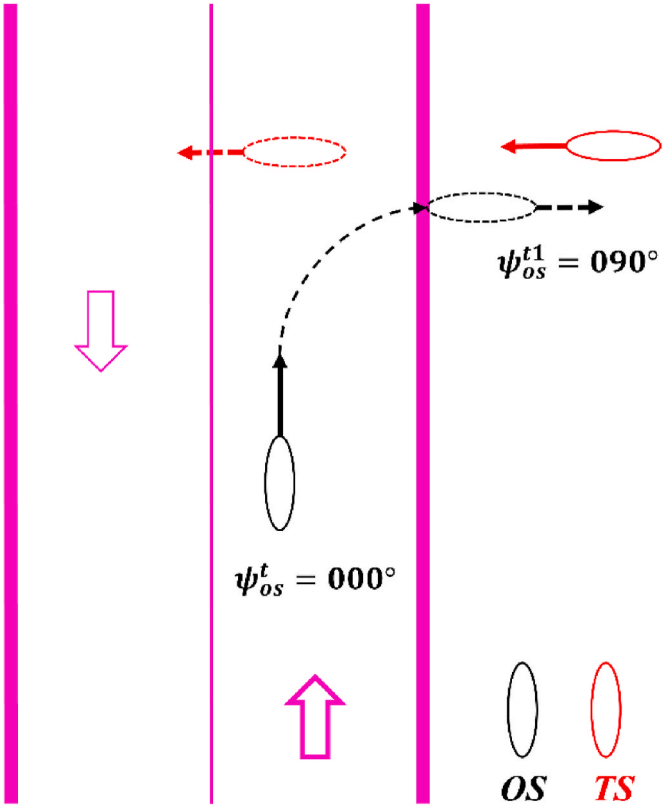


Fig. 2. Narrow waterway scenario diagram.

collision avoidance co-operation actions for stand-on ship.

The study by Lee and Furukawa (2022) was conducted using a MMG model, and the model ship was KVLCC2. MMG models can have errors in calculating ship trajectories (Lee and Furukawa, 2022). Furthermore, there are many types of ships other than VLCCs at sea. The maneuverability of each type of ship is different, so the timing of avoidance action may vary. Therefore, further research is needed to build on the work presented by Lee and Furukawa (2022).

This study will be conducted on the KCS using computational fluid dynamics (CFD), which can calculate the ship's course more accurately than the MMG model.

Table 1
The main properties of the KCS model used in this work.

Main particulars	Symbols	Model scale (1:75.24)	Full scale
Length between the perpendiculars	L_{BP} (m)	3.057	230.0
Length of waterline	L_{WL} (m)	3.0901	232.5
Beam at waterline	B_{WL} (m)	0.4280	32.2
Draft	D (m)	0.1435	10.8
Displacement	Δ (m^3)	0.1222	52030
Block coefficient	C_B	0.651	0.651
Ship wetted area without rudder	S (m^2)	1.6834	9530
Longitudinal centre of buoyancy	% L_{BP}, fwd+	−1.48	−1.48
Metacentric height	GM (m)	0.008	0.6
Radius of gyration	K_{xx}/B	0.49	0.49
Radius of gyration	K_{yy}/L_{BP}, K_{zz}/L_{BP}	0.25	0.25
Propeller diameter	D_P (m)	0.105	7.9
Propeller rotation direction (view from stern)		Right hand side	Right hand side
Rudder turn rate	(deg/s)	20.1	2.32

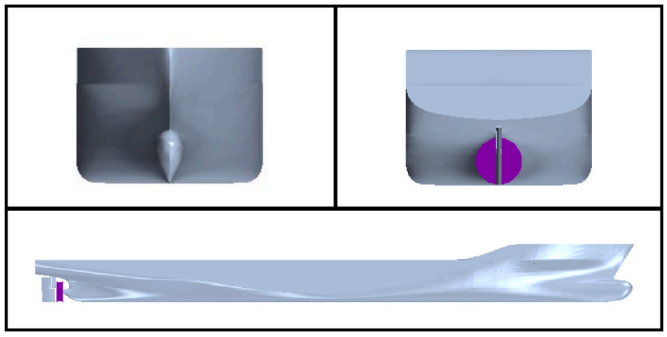


Fig. 3. KCS geometry with a semi balanced rudder and an actuator disk.

Researchers have investigated ship maneuvering performance using experimental methods, notably through model testing. Traditionally, free-running tests in model basins have offered reliable evaluations of maneuverability, but they come with certain drawbacks. Despite providing valuable insights into a ship's maneuvering characteristics, these tests are hindered by issues related to time, cost, and technical challenges. Additionally, conducting these experiments requires large-scale basin facilities, which are available only to a limited number of research institutions with the necessary resources (Sanada et al., 2013, 2019; Elshiekh, 2014; Kim et al., 2019; Yasukawa et al., 2021).

With the ongoing improvements in computational technology, CFD techniques have become a flexible tool for analyzing ship hydrodynamics. As computational power and numerical methods advance, CFD is increasingly capable of simulating complex, nonlinear ship maneuvers at sea, capturing crucial viscous and turbulent effects that influence maneuvering dynamics. Using CFD for maneuvering studies provides significant benefits, offering detailed insights into hydrodynamic forces, moments, surface interactions, and velocity/pressure distributions. These detailed results contribute to a more profound understanding of the complex hydrodynamic phenomena involved in ship maneuvers.

Musdscari et al. (2008) advanced the field by performing detailed free-running CFD simulations on a self-propelled twin-screw patrol vessel operating in deep, unrestricted waters. They employed a simplified propeller model to effectively capture the hydrodynamic forces and moments involved in the vessel's turning maneuvers. In a related study, Broglia et al. (2013) enhanced the understanding of ship maneuverability by examining the turning performance of a twin-screw, single-rudder model in calm, deep water. Their use of an Unsteady Reynolds-Averaged Navier-Stokes (URANS) solver, combined with the novel application of the actuator disk approach, yielded valuable

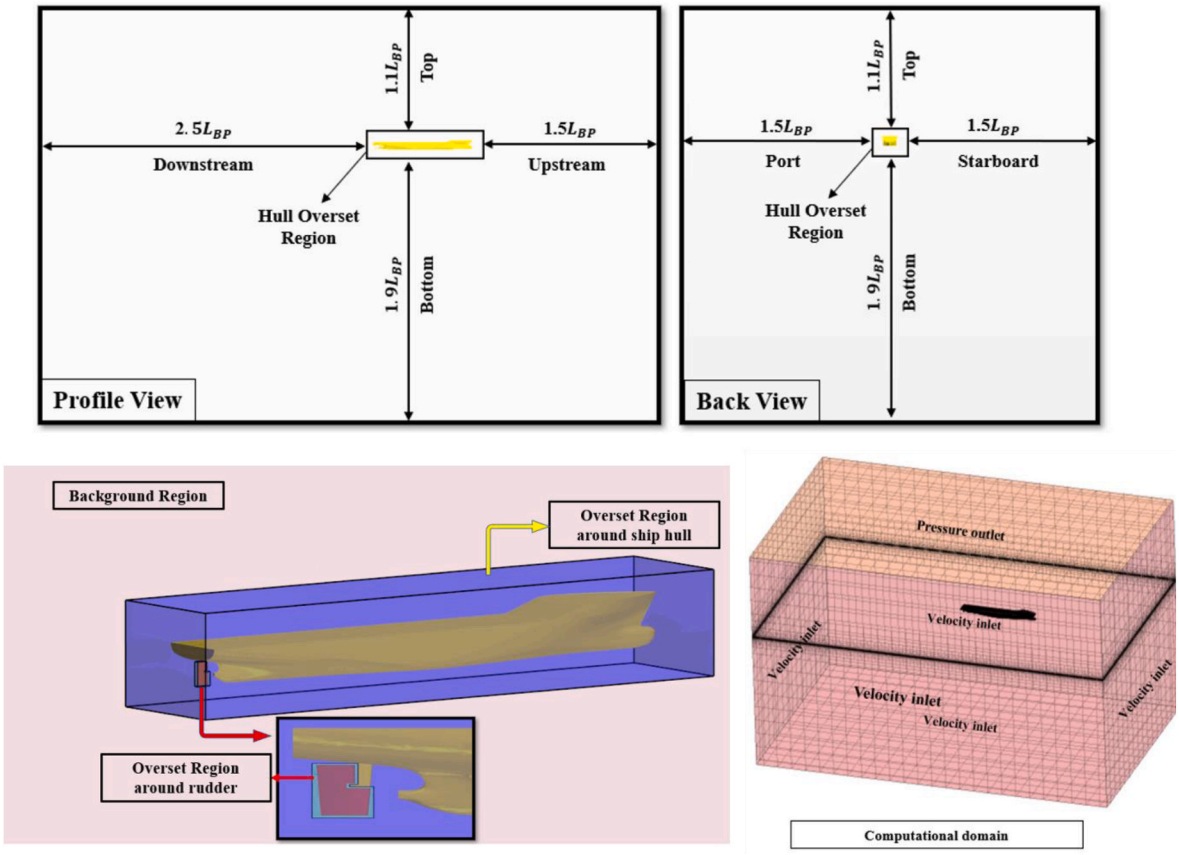


Fig. 4. Computational domain and imposed boundary conditions utilized in this work.

insights into the complex interactions between the ship's hull and its propeller.

Mofidi and Carrica (2014) made notable progress in validating free-running CFD simulations by using CFD Ship-low, a general-purpose CFD software developed at the University of Iowa. Their study focused on simulating the zigzag maneuver of the KCS model in calm water, comparing the numerical results with experimental data. Their findings demonstrated that the CFD simulations closely aligned with experimental observations, confirming the effectiveness and reliability of the free-running CFD approach for analyzing ship maneuvering dynamics. Following this work, several researchers have adopted similar methods to estimate ship maneuvering behavior in calm water (Broglia et al., 2015; Dubbioso et al., 2016; Hasanvand and Hajivand, 2019; Hasanvand et al., 2021; Kim et al., 2021c; Kinaci and Ozturk, 2022; Shen et al., 2015). These collective endeavors have consistently demonstrated the capability of free-running CFD simulations to accurately predict the maneuvering performance of ships.

Furthermore, recent studies have broadened the scope of free-running CFD simulations to evaluate ship maneuverability across diverse wave conditions (Kim et al., 2021a; Kim et al., 2021b; Kim et al., 2022a Kim et al., 2022b; Kim and Tezdogan, 2022). These simulations have played a crucial role in evaluating autonomous control performance, providing important insights that contribute to the development of future autonomous marine navigation systems (Kim et al., 2023a, 2023b). This extension of research not only enhances our understanding of ship maneuverability but also paves the way for the development of more efficient and reliable autonomous navigation technologies in maritime environments.

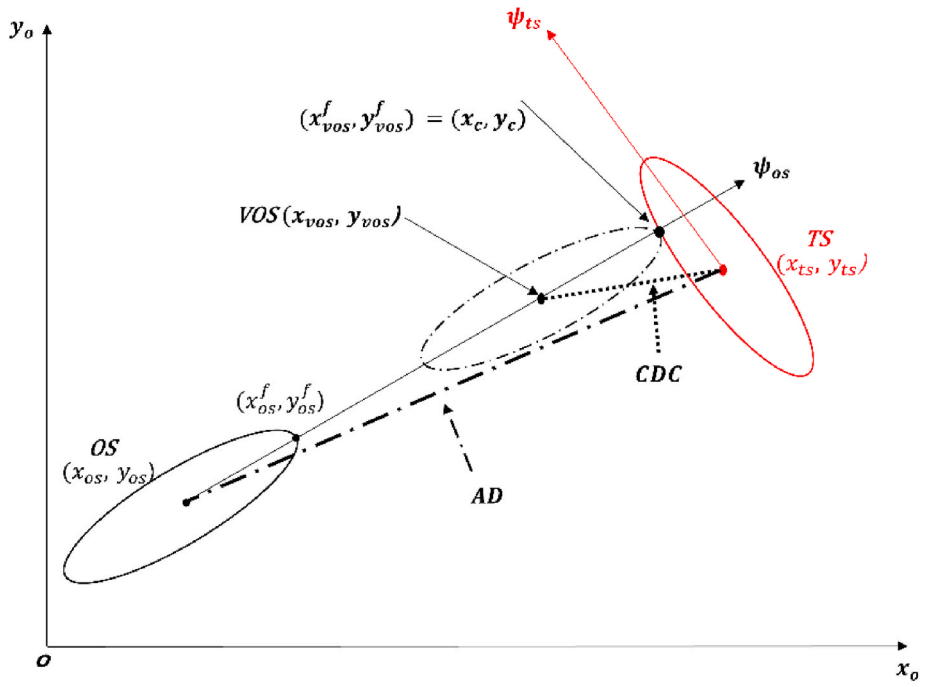
Therefore, based on the CFD-based trajectory of the KCS, this study will identify situations where collision of the KCS cannot be avoided and suggest the timing of collision avoidance action. Furthermore, we will analyze how collision avoidance timing changes with changes in ship's speed, and the timing of collision avoidance cooperative actions when KCS is the stand-on ship will be proposed.

Nomenclature

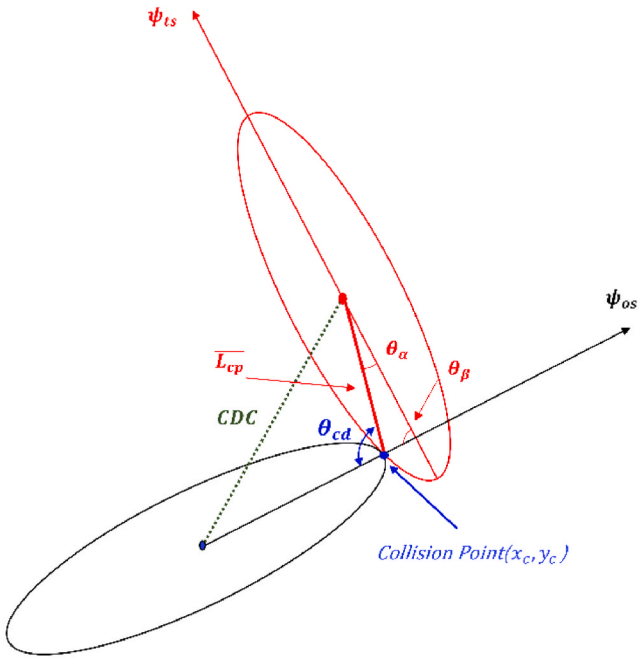
Symbol	Definition
x_{os}, y_{os}	Coordinates of the OS in an Earth-fixed coordinate system
x^f_{os}, y^f_{os}	Coordinates of the OS forecastle point in the Earth-fixed coordinate system
x_{vso}, y_{vos}	Coordinates of the OS in the Earth-fixed coordinate system
x^f_{vos}, y^f_{vos}	Coordinates of the VOS forecastle point in the Earth-fixed coordinate system
x_{ts}, y_{ts}	Coordinates of the TS in the Earth-fixed coordinate system
ψ_{os}, ψ_{ts}	Headings of the OS and RS
x_c, y_c	Coordinates of the collision point in the Earth-fixed coordinate system
θ_{ot}	Intersection heading angle between ψ_{os} and ψ_{ts}
CDC	Collision Distance at Collision
OS	Own ship
TS	Target ship
δ_{os}	Rudder angle of the OS
L	Ship length
AD	Actual distance between the OS and RS
θ_{ic}	θ_{Ot} at which collision is inevitable

2. Methodology

In this section, the methodology for deriving the timing of collision avoidance cooperation for stand-on ships is described. We will first clarify how the scenario was organized and how the trajectories of the ships were obtained, then proceed to explain the CDC that will be utilized to evaluate the collision between the two ships.



(a)



(b)

Fig. 5. CDC's calculations.

2.1. Configure a scenario

Similar to the study by Lee and Furukawa (2022), this study assumes that a collision can occur when two ships are very close together in a narrow waterway. Therefore, it is necessary to construct a scenario to identify areas where two ships are so close that collision cannot be avoided by the behavior of the give-way ship alone. For this purpose, we consider only the crossing situation where the intersection heading angle between the two ships (θ_{ot}) is from 2 ° to 90 °, as shown in Fig. 1.

In Fig. 1, the horizontal and vertical axes represent the x and y coordinates, respectively. The black and red ellipses represent the own ship (OS) and the target ship (TS), while the red curved dotted line indicates the initial position of the TS. The angle θ_{ot} represents the heading angle between the two ships.

The maximum value of θ_{ot} is set to be 90 ° because in the worst-case scenario in a narrow waterway, a give-way ship must execute an

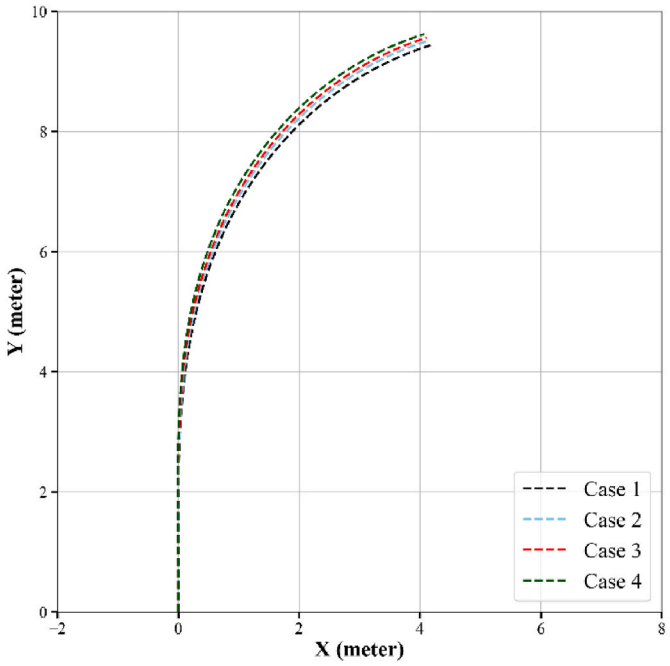


Fig. 6. Ship's trajectory.

Table 2
Ship's coordinates at the end of the simulation.

	Time	x	y
Case1	15.67s	4.18	9.43
	13.86s	4.12	9.50
Case2			
	12.43s	4.12	9.56
Case3			
	11.20s	4.07	9.62
Case4			

Table 3
Distance difference in each case at the end of the calculation.

		x	y	Δdistance
Case1	None	–	–	–
Case2	Case2 – Case1	– 0.06m	0.07m	0.09 (0.03L)
Case3	Case3 – Case1	– 0.06m	0.13m	0.14 (0.05L)
Case4	Case4 – Case1	– 0.11m	0.19m	0.22 (0.08L)

avoidance action involving a 90 ° heading change to avoid colliding with a stand-on ship, as illustrated in Fig. 2.

In the presented scenario, the give-way ship is designated as OS and the stand-on ship is designated as TS. Only the give-way ship in Fig. 1 utilizes the rudder for collision avoidance action. The rudder is used to calculate the ship's trajectory, with a maximum angle of 35 ° applied for emergency action. The speeds of both ships are set to be the same in this study, and the model ships are of identical size. The model ship in this study is a KCS (container ship), which has a different C_b and different maneuvering performance compared to the model ship of KVLCC2 in the previous study by Lee and Furukawa (2022). Since the turning ability of container ships is generally better than that of VLCC (Lee et al., 2021), it is expected that results will be different from previous studies.

In addition, in this study, the speed of the model ship was differentiated into four cases, 14.5 kts, 16.5 kts, 18.5 kts, and 20.5 kts, to determine the angle at which a collision becomes inevitable based on the ship's speed. The initial speed of 14.5 kts was chosen because it is considered unrealistic to have a speed lower than 14.5 kts, and the speed for CFD validation is 14.5 kts, which is also considered realistic for container carrier. Of course, this does not mean that CFD is inaccurate except at 14.5 kts. CFD is capable of calculating accurate ship's trajectory in all speed ranges. And we wanted to identify what happens when a container carrier sails at high speed in this study, so we conducted the study by increasing the speed by 2 kts from the initial speed. The parameters of the model ship used in this study are detailed in Table 1 and Fig. 3.

2.2. Calculation of ship trajectory

Recent advancements in CFD technology, along with enhanced computational capabilities, have established CFD simulations as an attractive alternative to conventional experimental methods. Free-running CFD simulations are increasingly acknowledged as an advanced numerical approach for evaluating ship maneuvering performance, reflecting a notable shift in research methodologies. Various studies utilizing CFD simulations to investigate ship maneuverability have effectively showcased the utility of free-running CFD in analyzing maneuvering dynamics, as evidenced by the research conducted by (Broglia et al., 2015; Dubbioso et al., 2016; Wang et al., 2016; Kim et al., 2021; Kim et al., 2021a; Kim et al., 2021b; Kim et al., 2021c). CFD is notable for its capacity to account for both viscous and rotational effects in fluid flow, providing more accurate maneuvering results compared to theoretical methods. Consequently, this study employed the advanced numerical approach of free-running CFD simulations to estimate the trajectory of a ship in operational sea conditions.

In this study, we conducted free-running simulations utilizing the commercial CFD package STAR-CCM+, version 15.04. While a concise overview of the primary features of the numerical modeling and computational methodology for the free-running CFD model is presented here, for a more comprehensive understanding, readers are directed to (Broglia et al., 2015; Dubbioso et al., 2016; Kim et al., 2022a, 2022b, 2022c, 2022d). The CFD code used in this study solves the unsteady Reynolds-Averaged Navier-Stokes (URANS) equations to analyze the turbulent flow around a ship performing a self-propelled maneuver. The conservation equations for mass and momentum in unsteady incompressible flows are expressed in vector form as follows:

$$\nabla \cdot \mathbf{U} = 0 \tag{1}$$

$$\frac{\partial(\rho \mathbf{U})}{\partial t} + \nabla \cdot \left[\rho(\mathbf{U} - \mathbf{U_g})\mathbf{U}\right] = -\nabla p + \nabla \cdot \left(\mu_{eff} \nabla \mathbf{U}\right) + \nabla \mathbf{U} \cdot \nabla \mu_{eff} + q_i \tag{2}$$

$\nabla \cdot (\cdot)$ and $\nabla (\cdot)$ stand for the divergence and gradient operators, respectively. $\mathbf{U}$ denotes fluid velocity and $\mathbf{U_g}$ denotes grid velocity; p denotes static pressure; ρ denotes fluid density; $\mu_{eff} = \rho(\nu + \nu_t)$ is effective dynamic viscosity, where ν and ν_t stand for the kinematic and eddy viscosity, respectively (ν_t is obtained from the turbulence model); q_i is a user-defined source term.

Utilizing a finite volume approach, the solver discretizes the governing equations and incorporates Menter's Shear Stress Transport (SST) model for system closure. The SST turbulence model uses a two-equation eddy viscosity approach, which integrates the k−ω turbulence model introduced by Wilcox specifically for the inner boundary layer region. As the flow progresses into the far-field, the model transitions smoothly to the conventional k-ε turbulence model. This transition is managed through blending functions based on the turbulence length scale. The SST turbulence model is well-regarded and frequently used in free-running CFD simulations found in the open literature due to its effectiveness and widespread application. (Kim and Tezdogan, 2022; Kim et al., 2023a, 2023b). The initial closure equation (Eq. (3)) represents the transport equation governing turbulent kinetic energy (k), and the subsequent equation (Eq. (4)) pertains to the transport of the specific dissipation rate (ω). The formulation of the SST turbulence model is expressed as follows:

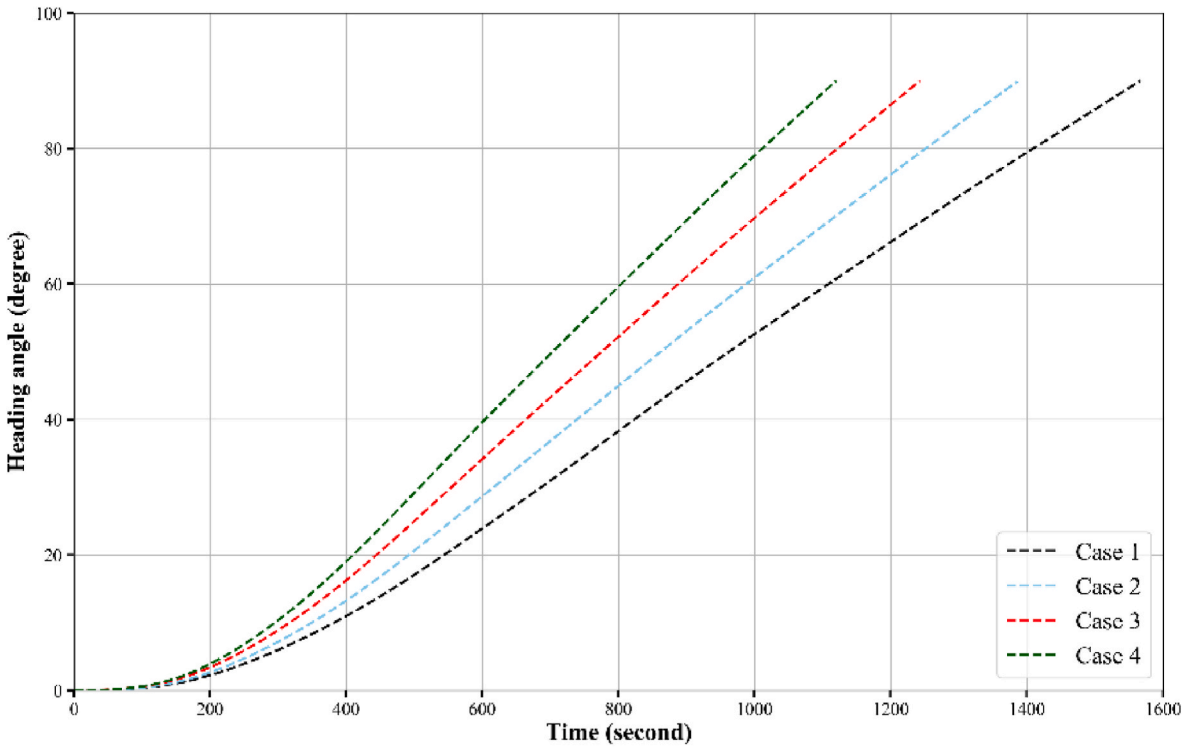


Fig. 7. Degree of heading change.

$$\frac{\partial(\rho k)}{\partial t}+\nabla\cdot(\rho k\boldsymbol{U})=\left[\mu_t\left(\nabla\boldsymbol{U}+(\nabla\boldsymbol{U})^T-\frac{2}{3}\nabla\cdot\boldsymbol{U}\right)-\frac{2}{3}\rho k\delta_{ij}\right]\nabla\boldsymbol{U} - \beta^*\rho\omega\boldsymbol{k}+\nabla\cdot[(\mu+\sigma_k\mu_t)\nabla\boldsymbol{k}] \tag{3}$$

$$\frac{\partial(\rho\omega)}{\partial t}+\nabla\cdot(\rho\omega\boldsymbol{U})=\frac{\gamma}{\nu_t}\left[\mu_t\left(\nabla\boldsymbol{U}+(\nabla\boldsymbol{U})^T-\frac{2}{3}\nabla\cdot\boldsymbol{U}\right)-\frac{2}{3}\rho k\delta_{ij}\right]\nabla\boldsymbol{U} - \beta\rho\omega^2+\nabla\cdot[(\mu+\sigma_\omega\mu_t)\nabla\omega]+2\rho(1-F_1)\frac{\sigma_{\omega2}}{\omega}\nabla\boldsymbol{k}\cdot\nabla\omega \tag{4}$$

In the SST turbulence model, blending functions transition between the k−ω and k-ε models based on flow conditions. Specifically, F_1 and F_2 are defined using tanh functions, with coefficients and constants from Menter (1994) determining the model's accuracy and closure.

The numerical simulations employed the SIMPLE algorithm to couple pressure and velocity. For modeling the multiphase flow of air and water, the volume of fluid (VOF) method was used. The impact of the rotating propeller during ship maneuvers was simulated using an actuator disk model, which applied axial and tangential body forces within the disk to represent its effects throughout the flow field.

Grid generation was accomplished using STAR-CCM+'s integrated meshing capabilities, creating a computational mesh with around 5.4 million cells. The domain was divided into three distinct regions: 1) the background region, 2) the hull overset region, and 3) the rudder blade overset region. A dynamic overlapping grid strategy was applied to handle significant ship and rudder movements during free-running maneuvers. The CFD model utilized an all-y + wall treatment approach, combining low-y + treatment for finer grids (where y+ < 5) and high-y + treatment for coarser grids (where y+ > 30). In this study, the average y + value on the hull surface is above 30. Fig. 4 presents the computational domain and boundary conditions utilized, as confirmed by previous research conducted by the authors (Kim et al., 2021a, 2021c).

Fig. 4 Schematic views of the computational domain applied for the simulations, as adapted in (Kim et al., 2021c).

The Dynamic Fluid Body Interaction (DFBI) method was used to accurately model the ship's movement. In the CFD model, the DFBI module allows for the simulation of the ship's motion with 6 degrees of freedom, enabling an evaluation of its maneuvering performance in calm water. This 6 ° of freedom motion solver computes the fluid forces, moments, and gravitational forces affecting the ship, and updates its position by solving the equations of motion. The specific equations for this motion are as follows:

$$\frac{\mathrm{d}(m\mathbf{V})}{dt}=\mathbf{f} \tag{5}$$

$$\mathrm{M_b}\frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{dt}}+\boldsymbol{\omega}\times\mathrm{M_b}\,\boldsymbol{\omega}=\mathbf{m} \tag{6}$$

where m is the mass of body, $\mathbf{V}$ is the velocity vector of its centre of mass, $\mathbf{f}$ is the resultant force vector acting on the ship, $\mathrm{M_b}$ indicates the tensor of moments of inertia in the body-fixed coordinate system, $\boldsymbol{\omega}$ is the angular velocity vector of the body and $\mathbf{m}$ denotes the resultant moment vector acting on the body.

2.3. CDC

CDC is a ship CA model presented by Lee and Furukawa (2022) that calculates the collision distance at all angles at which two ships can collide, compares it to the actual distance between two ships, and assesses whether and when a collision will happen. The accuracy of the model has been verified in some studies (Lee and Furukawa, 2022; Lee and Park, 2024). The calculation process of CDC is as follows.

The black ellipse is the OS, the black dotted ellipse is the virtual OS (VOS), and the red ellipse is the TS in Fig. 5(a). In an earth-fixed coordinate system, the reference point of the OS is denoted by x_{os}, y_{os}, the reference point of the TS is denoted by x_{ts}, y_{ts}, and their headings are denoted by ψ_{os} and ψ_{ts}, respectively. The reference point of the VOS is denoted by x_{vos}, y_{vos}, and the coordinates of the OS and VOS headings are denoted by $x^f_{os}, y^f_{os}, x^f_{vos}$, and y^f_{vos} as shown in Fig. 5(a).

The coordinates of the point where the two ships collide are x_c, y_c in Fig. 5(b). Then, the CDC is defined as the distance between the two ships

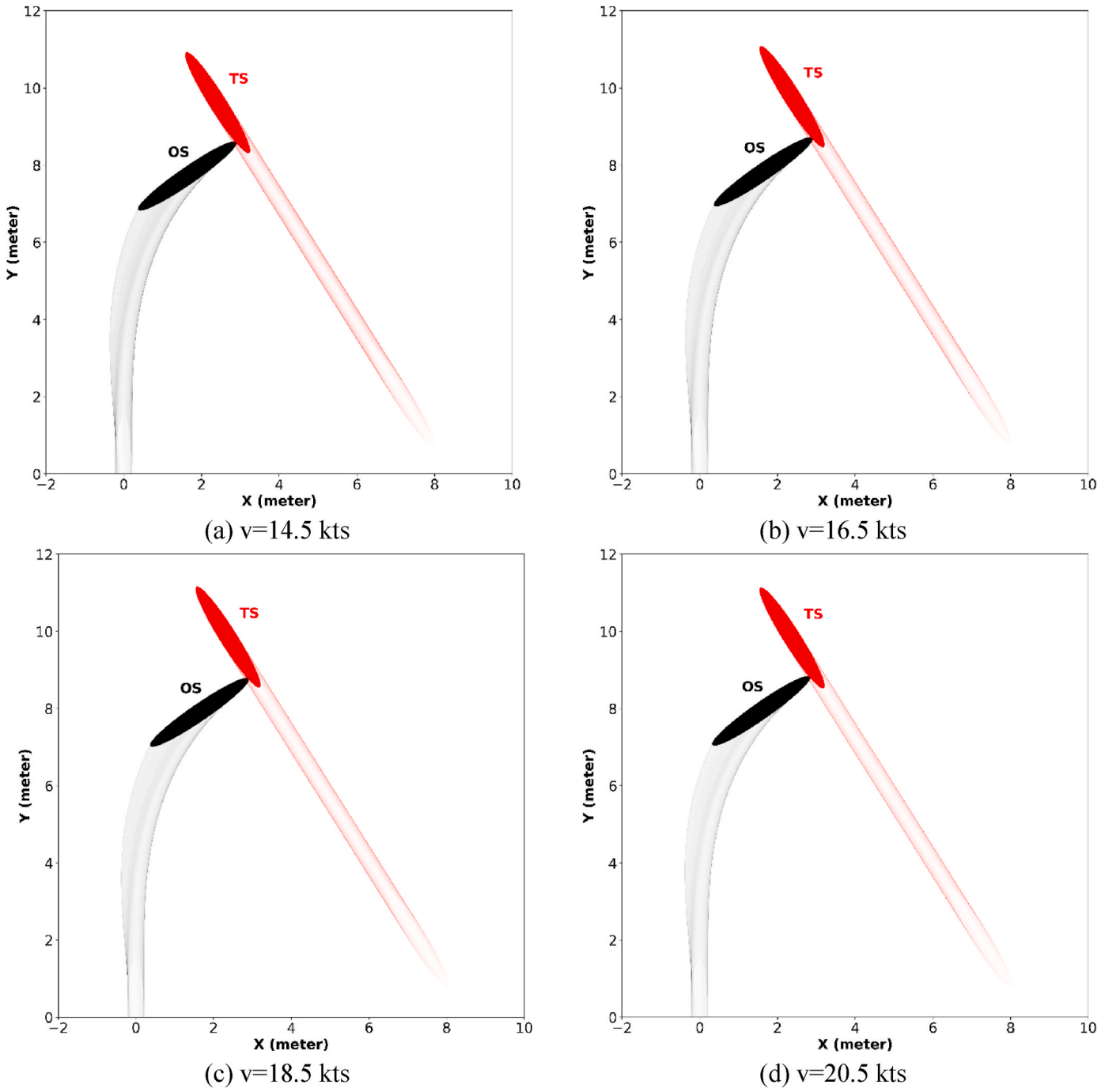


Fig. 8. Ships' trajectory at maximum $\theta_{ic} = 32^{\circ}$ by speed.

at the time of collision, and AD is the actual distance between the two ships. To calculate the CDC, the OS is moved to a virtual position without changing its heading so that it is in contact with the TS. The heading of the OS and the ellipse of the TS will then be tangential (i.e., the extended centerline of the OS will be tangential to the ellipse of the TS). The ellipse equation for the TS is given by

$$\frac{\{(y_o - y_{ts})\cos\psi_{ts} + (x_o - x_{ts})\sin\psi_{ts}\}^2}{(L/2)^2} + \frac{\{-(y_o - y_{ts})\sin\psi_{ts} + (x_o - x_{ts})\cos\psi_{ts}\}^2}{(B/2)^2} = 1 \tag{10}$$

The extended centerline of the OS is given by

$$y_o = (x_o - x_{os})\tan\psi_{os} + y_{os} \tag{11}$$

Substituting Eq. (11) into Eq. (10) obtains

$$Ax_o^2 + 2Bx_o + C = 0 \tag{12}$$

Table 4
CA result.

Speed	14.5 kts	16.5 kts	18.5 kts	20.5 kts
Collision time (s)	10.38	9.29	8.36	7.52
CDC (m)	2.057	2.083	2.087	2.047
AD (m)	2.054	2.080	2.085	2.044
AD-CDC (m)	−0.003	−0.003	−0.002	−0.002
CA result	Collision	Collision	Collision	Collision

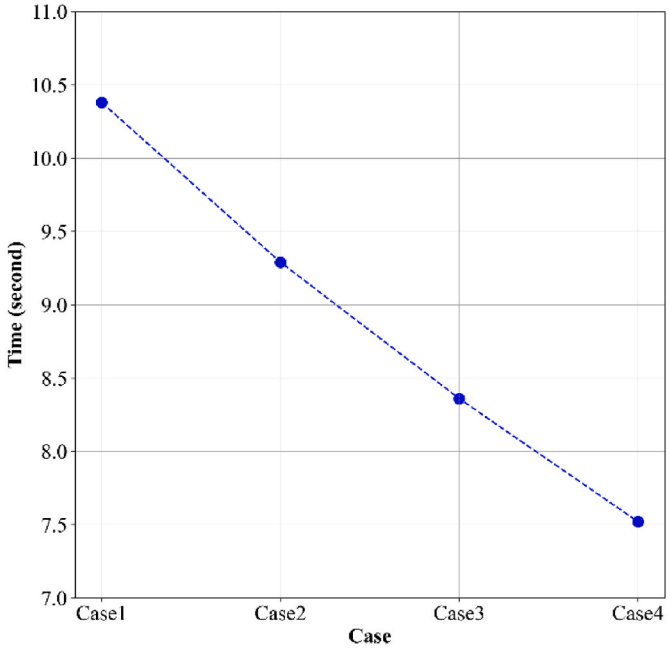


Fig. 9. Collision time.

Then, x_o can be obtained

$$x_o = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \quad (x_o \in x_c) \tag{13}$$

We then define the following terms:

$$\left.\begin{aligned} L_{cp} &= \sqrt{(x_c - x_{ts})^2 + (y_c - y_{ts})^2}, \\ \theta_{cd} &= \psi_{os} + (2\pi - \psi_{ts}) + \tan^{-1}\left(\frac{x_c - x_{ts}}{y_c - y_{ts}}\right). \end{aligned}\right\} \tag{14}$$

These terms can then be used to calculate the CDC:

$$CDC = \sqrt{\left(\frac{L_{OS}}{2}\right)^2 + (L_{cp})^2 - 2\left(\frac{L_{OS}}{2}\right)L_{cp}\cos\theta_{cd}} \tag{15}$$

The AD can be obtained by using the reference points of the OS and RS:

$$AD = \sqrt{(x_{os} - x_{ts})^2 + (y_{os} - y_{ts})^2}$$

3. Result

In the previous section, we described how the experiments in this study are designed, how the ship's trajectory is calculated, and how CA is performed. In this section, we will analyze the results of CA with CDC to provide timing of stand-on ship's cooperative avoidance action and verify reliability of CDC model.

3.1. Ship trajectory

It is easy to predict that the trajectory of the ship will differ according to speed. To analyze how trajectories change, we first plotted the trajectories of ships according to speed differences in Fig. 6.

In Fig. 6, the horizontal and vertical axes represent the x- and y-coordinates, and the black, sky blue, red, and green dotted lines represent speeds from 14.5 kts to 20.5 kts denoted as cases 1 to 4, respectively. As shown in Fig. 6, we can see that the trajectory of the ship varies depending on the speed. However, the differences are not significant, and the distance difference (d_f) based on 14.5 kts is shown in Tables 2 and 3.

Table 2 shows the x- and y-coordinates and time when the ship's heading reaches 90 °, and Table 3 shows the distance difference in meters and L value at the end of the ship's trajectory calculation. As can be seen in Table 3, there is a difference in distance, but the difference in the ship's length (L) value is insignificant.

The simulation observed that differences in speed does result in variations in the ship's trajectory; however, a significant difference was not noted. Nevertheless, the degree of heading change did vary to a certain extent in accordance with speed. Fig. 7 shows the heading change in each case.

In Fig. 7, the horizontal axis represents time, and the vertical axis represents the heading of the OS. As shown in Fig. 7, in all cases, the ship's heading reached 90 °, but the time taken to do so ($\psi_{os}^t = 90°$) varied. The amount of heading change was greater at higher speeds. This is because the turning momentum is bigger at higher speeds. Case 1 took approximately less than 16.00 s, and the fastest Case 4 was approximately 11.00 s.

Combining Figs. 6 and 7, we can see that the faster the speed, the slightly larger the diameter of the ship's trajectory and the faster the time to reach $\psi_{os} = 90°$, by up to 5 s.

3.2. Collision assessment

Fig. 8 shows the trajectory of the two ships in intersection heading (θ_{ic}) where the collision inevitably occurred, and Table 4 presents a detailed CA result for each case of θ_{ic}.

In Fig. 8, the horizontal and vertical axes represent the x- and y-coordinates in meters, with the black ellipse representing the OS and the red ellipse representing the TS. As shown in Fig. 8, the two ships collided at the stern end of the TS. Fig. 8 is insufficient for a detailed analysis of the collision between two ships; therefore, we will analyze it by examining the CDC's calculated values in the context of the collision.

As shown in Table 4, collisions between the two ships happened in every case. The situation at the time of the collision is characterized by the AD being smaller than the CDC. As indicated in Table 4, the AD is smaller by 0.002 m–0.003 m in every case, which confirms that a collision occurred.

Fig. 9 shows the time of collision in each case. In Fig. 9, the horizontal axis represents the case, and the vertical axis represents the time of collision. As shown in Fig. 9 and Table 4, the collision occurred at different times due to varying speeds. Case 1 is the slowest, with a collision time of 10.38 s, whereas Case 4 is the fastest, with a collision time of 7.52 s. The difference in time between these two cases is about 2.86 s, indicating that reducing the speed in Case 1 could delay the collision time by approximately 38%. Additionally, the distance between the two ships at the time of collision was about 2.044 m–2.085 m, demonstrating that the collision occurred when the distance between the ships was not zero.

What is interesting is that θ_{ic} does not change with different speeds of the ship, only the time of the collision was changed. This raises the challenging question of whether to reduce or maintain speed in the event of an imminent collision. A slow-speed ship may be less maneuverable, while a high-speed ship can generate a considerable amount of collision energy in a collision. It is challenging to determine the optimal

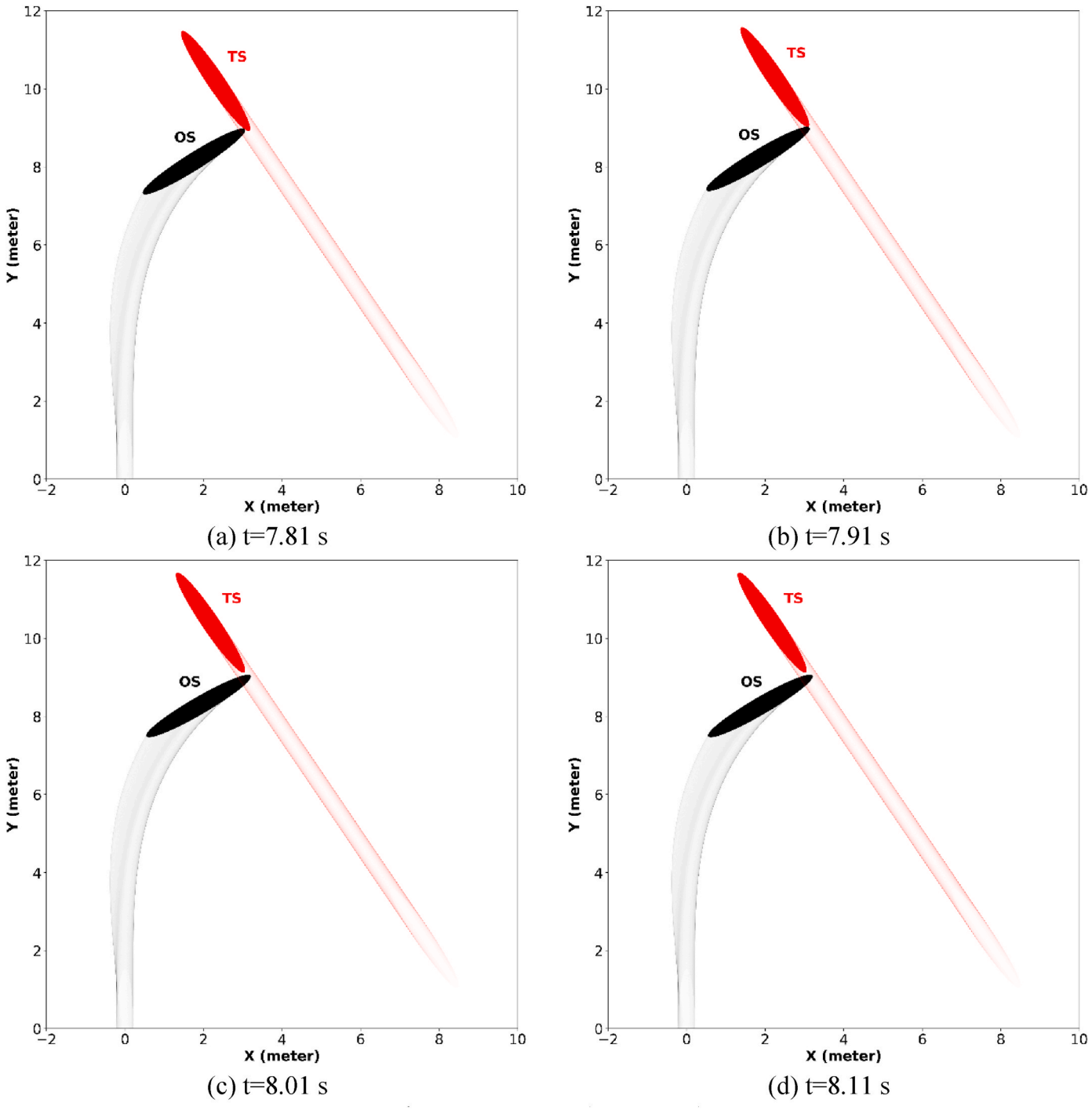


Fig. 10. Case 4 ($\theta_{ot} = 34°$).

speed to maintain in the event of an unavoidable collision.

This study is unable to provide a definitive solution to the aforementioned question. This is due to the fact that this study only considered ship speeds between 14.5 kts and 20.5 kts. Nevertheless, based on the findings of this study, it can be proposed that in the event of an unavoidable collision between two ships, reducing speed to 14.5 kts may be advantageous in postponing the collision.

3.3. Verification of CDC reliability

The CA was performed based on the CDC calculation in which the two ships were in very close each other. Therefore, while the reliability of CDC has been confirmed in previous studies, it also requires to be verified in this study.

Fig. 10 shows the closest passing of the two ships without collision over time in this study. In Fig. 10, the horizontal and vertical axes represent the *x*- and *y*-coordinates in meters, and the black and red ellipses represent the OS and TS, respectively. In Fig. 10(a) and (b), the two ships appear to be on a collision each other, and then they seem to pass each other. Although CA results show no collision occurred, it may seem like a collision in Fig. 10(a) and (b).

Fig. 11 shows a zoomed-in view of what appears to be a collision between two ships in Case 4, with $\theta_{ot} = 34°$. In Fig. 11, the horizontal and vertical axes represent the *x*- and *y*-coordinates in meters, and the

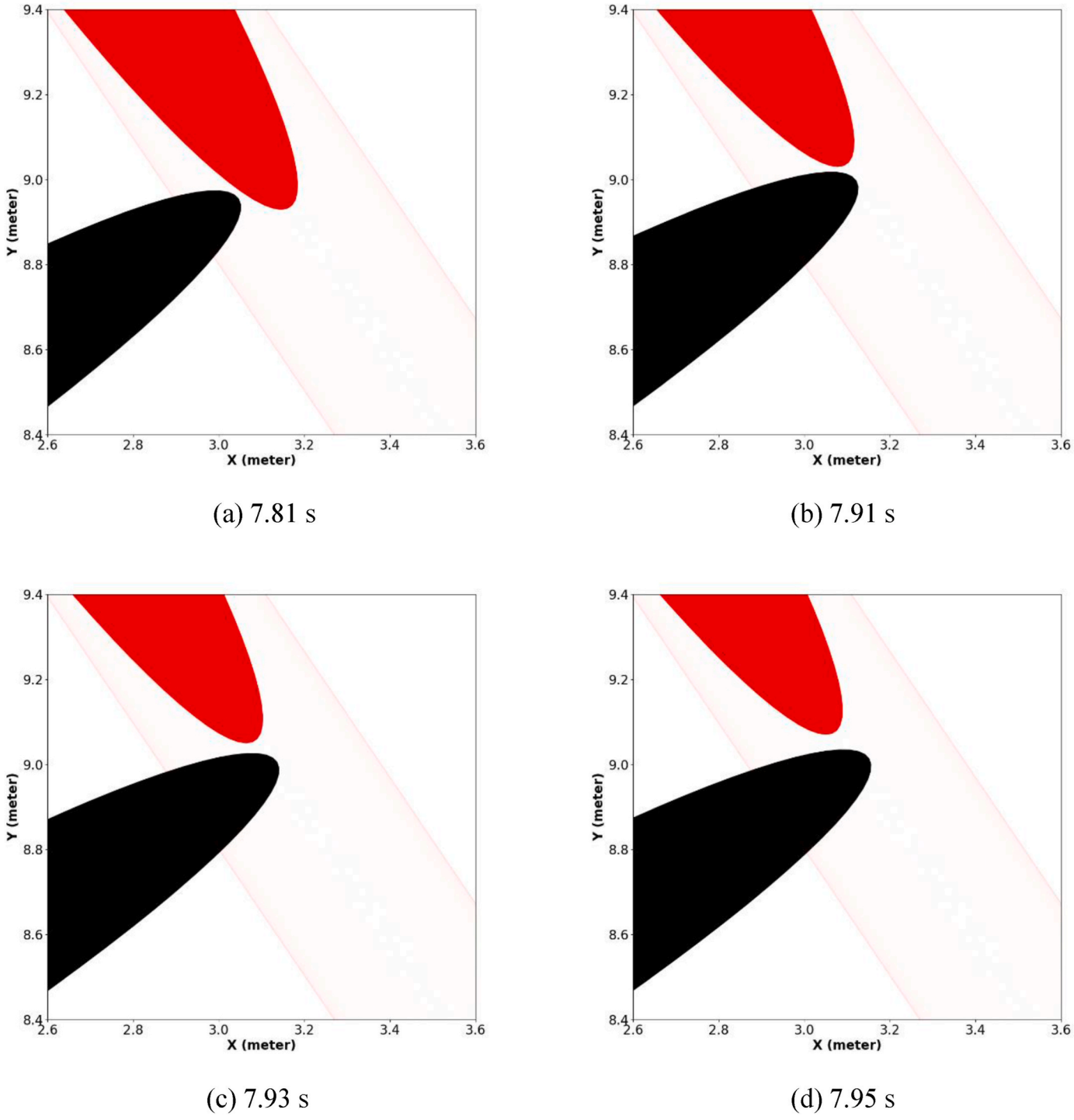


Fig. 11. Zoom in Fig. 10.

Table 5
Detailed calculation of Case 4 ($\theta_{ot} = 34^{\circ}, v = 20.5$ kts).

Time	7.81 s	7.82 s	7.83 s	7.84 s	7.85 s	7.86 s	7.87 s	7.91 s
CDC (m)	2.124	2.126	2.129	2.130	2.131	2.130	2.127	N/A
AD (m)	2.138	2.138	2.138	2.138	2.139	2.139	2.140	2.143
AD−CDC(m)	0.014	0.011	0.001	0.008	0.007	0.009	0.013	N/A
CA result	NC	NC	NC	NC	NC	NC	NC	NC

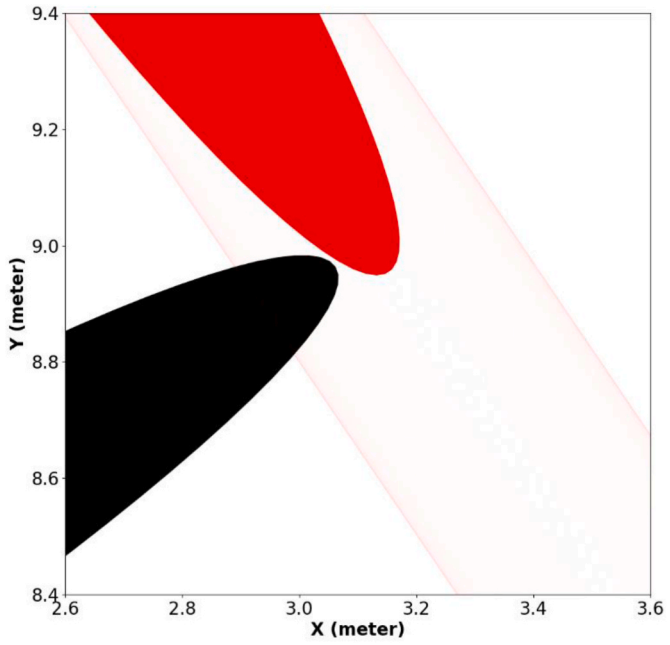


Fig. 12. Zoom in case 4 ($\theta_{ot} = 34°$, at 7.83s).

black and red ellipses denote the OS and TS, respectively. As shown in Fig. 11 collision and any contact have not happen between the two ships. To analyze this with detailed calculated values, we also present the CA result in Table 5.

Table 5 shows the CA result in specific simulation times. We can see that no collision occurred in Fig. 11(a) and (b), and the CDC calculation confirms that no collision took place as shown in Table 5. Notably, Table 5 indicates that the closest approach was 0.001 m at 7.83 s. This is also illustrated for detail in Fig. 12.

Fig. 12 shows the position of the two ships such as the OS and the TS in Case 4, $\theta_{ot} = 34°(t = 7.83s)$. It depicts the closest moment between the two ships' approach in this study. The distance between the two ships is 0.001 m, verifying that no collision occurred. This confirms the reliability of CA result using CDC.

3.4. Summary

Prior to this study, Lee and Furukawa (2022) conducted a similar study using KVCC2. In their study, the timing was within 17.5 s, θ_{ic} was 34 °, and the ship's trajectory was calculated by the MMG model. In this study, the KCS was selected as a model ship, which has better turning ability compared to KVLCC2, which has difference in C_b. Although the comparison of timing between KCS and KVLCC2 may not be possible exactly due to the different scales of the model ships, it is clear that turning abilities can be different θ_{ic} due to different ship's maneuverability. Therefore, the action timing for collision avoidance cooperation should be also considered according to the ship's type. This study suggests that the timing of collision avoidance cooperation actions for the stand-on ship, as outlined in Rule 17 of COLREGS, may vary depending on the ship type. Of course, various factors, such as sea conditions, may affect the timing of collision avoidance actions, but this remains a topic for further research.

Fig. 13 compares the results of Lee and Furukawa (2022) with those from this study. In Fig. 13, the horizontal and vertical axes represent the *x*- and *y*-coordinates, and the black ellipse denotes the OS and the red dotted ellipse represents the TS. As illustrated in Fig. 13, θ_{ic} values for the two ships differ, and Fig. 13(b) shows that the initial position of TS can vary due to the different speeds of the ships.

Eq. (16) below summarizes the results of both studies.

$$T_c = \sum f(M, V_{OT}, D_{OT}, \Theta_{OT}) \tag{16}$$

In Eq. (16), T_c is the time of collision, M is the maneuvering performance of the ship, V_{OT} is the difference in speed between OS and TS, and Θ_{OT} is the intersection heading angle of the two ships. And Eq. (17) is a more detailed representation of Eq. (16).

$$\left.\begin{array}{l} M = t(\psi_{os} = 90°) \\ V_{OT} = v_{os} - v_{ts} \\ D_{OT} = PS_{os} - PS_{ts}\ (t = 0\ s) \\ \Theta_{OT} = 2\pi - (\psi_{ts} - \psi_{os}) \end{array}\right\} \tag{17}$$

4. Discussion

This study used KCS to identify situations where collisions are inevitable. We predicted that the results would differ significantly from

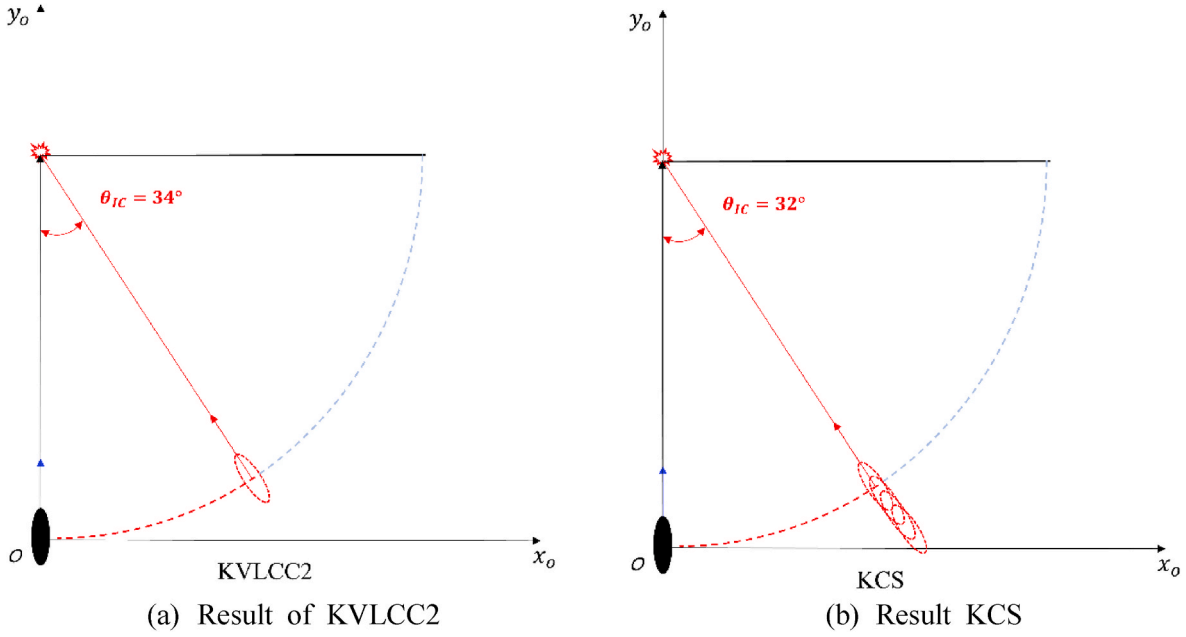


Fig. 13. Research findings from KVLCC2 and KCS.

previous studies. This is because the previous study was conducted with KVLCC2, which has relatively poor turning ability. They found a $\theta_{ic} = 34^{\circ}$, while this study found $\theta_{ic} = 32^{\circ}$ for KCS, this result shows that difference of θ_{ic} between KCS and KVLCC2 is not significant.

Although difference of θ_{ic} is small, there is bound to be a difference in the time when the collision occurs. We could confirm that by changing ship speed from 14.5 kts to 20.5 kts. And it was verified that θ_{ic} did not change depending on the ship's speed in this study.

If a collision is imminent, we know from experience that we should reduce ship speed, but it is a difficult decision because reducing the ship's speed too much can reduce the ship's ability to turn. This means that it is difficult to determine how much to reduce the speed of the ship. It can be said from this study result that it is more beneficial to lower the speed by 14.5 kts when two ships are very close each other, as it can delay the time of collision.

Nevertheless, this study is limited by the fact that the speeds of the two ships were the same. Also, since this is the first study on KCS using CFD, there is a limitation because only one model ship was selected in this study, and like Lee and Furukawa (2022), sea conditions were not considered. This should be addressed in future studies.

5. Conclusions

This study aims to present timing of collision avoidance cooperation action of a stand-on ship. For this purpose, KCS was designated as a model ship, and the trajectory of the ship was obtained through CFD. The speed of the ship increased by 2 kts, from 14.5 kts to 20.5 kts, and CA was performed using the CDC. This study verified that, although the amount of heading change is faster at higher speeds, the time to collision is also shorter between the two ships. At 20.5 kts, the collision occurred after 7.52 s, whereas at 14.5 kts, the collision occurred after 10.38 s. It would be advantageous to reduce the ship's speed to delay the time of collision as much as possible in a narrow waterway in order to find actions that can avoid the collision. Therefore, in some narrow waterways, speed limits should be considered in areas of heavy ship traffic, and a reduction of ship speed will give ships more time to avoid collisions in very dangerous situations. Although the collision time varied in each case, θ_{ic} was surprisingly the same at 32° in every case. These results are due to the short simulation time, which causes the ship's trajectory to differ little while being simulated.

Accordingly, the timing of the collision avoidance cooperative action of the stand-on ship in accordance with the speed in KCS is presented in Table 4. The results demonstrate that when θ_{ic} is less than 32°, the stand-on ship should engage in collision avoidance cooperative action when the speed of the two ships is 14.5 kts and a collision is expected after 10.38 s. Conversely, if the speed between the two ships is 20.5 kts, the stand-on ship should take a collision avoidance cooperative action if a collision is expected in 7.52 s.

This study provides concrete verification of Lee and Furukawa (2022) and Lee and Park (2024), demonstrating that the timing of the stand-on ship's collision avoidance cooperation action can vary depending on the ship's maneuvering performance, ship's speed and different types of ships. Of course, the trajectory of the ship may change depending on sea conditions, which were not considered in this study. Although sea conditions do not significantly affect the ship's trajectory during the short simulation time and do not impact the change in θ_{ic} (Lee and Park, 2024), this aspect will be also carefully considered in future studies. Therefore, we plan to use KCS to study how the ship's trajectory changes under various sea conditions and how θ_{ic} changes with the change in ship's trajectory.

CRediT authorship contribution statement

Hee-Jin Lee: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Daejeong Kim: Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Broglia, R., Dubbioso, G., Durante, D., Di Mascio, A., 2013. Simulation of turning circle by CFD: analysis of different propeller models and their effect on manoeuvring prediction. Appl. Ocean Res. 39, 1–10.
Broglia, R., Dubbioso, G., Durante, D., Di Mascio, A., 2015. Turning ability analysis of a fully appended twin screw vessel by CFD. Part I: single rudder configuration. Ocean Eng. 105, 275–286.
Chun, D.H., Roh, M.I., Lee, H.W., Ha, J., Yu, D., 2021. Deep reinforcement learning based collision avoidance for an autonomous ship. Ocean Eng. 234, 109216.
Du, L., Goerlandt, F., Banda, O.A.V., Huang, Y., Wen, Y., Kujala, P., 2020a. Improving stand-on ship's situational awareness by estimating the intention of the give-way ship. Ocean. Eng. 201, 107110.
Du, L., Banda, O.A.V., Goerlandt, F., Huang, Y., Kujala, P., 2020b. A COLREG-compliant ship collision alert system for stand-on vessels. Ocean. Eng. 218, 107866.
Dubbioso, G., Durante, D., Di Mascio, A., Broglia, R., 2016. Turning ability analysis of a fully appended twin screw vessel by CFD. Part II: single vs. twin rudder configuration. Ocean Eng. 117, 259–271.
Elshiekh, H.A., 2014. Maneuvering characteristics in calm water and regular waves for ONR Tumblehome. The University of Iowa.
Gao, M., Shi, G.Y., 2020. Ship collision avoidance anthropomorphic decision-making for structured learning based on AIS with Seq-CGAN. Ocean Eng. 217, 107922.
Hasanvand, A., Hajivand, A., 2019. Investigating the effect of rudder profile on 6DOF ship turning performance. Appl. Ocean Res. 92, 101918.
Hasanvand, A., Hajivand, A., Ali, N.A., 2021. Investigating the effect of rudder profile on 6DOF ship course-changing performance. Appl. Ocean Res. 117, 102944.
Hasegawa, K., Fukuto, J., Miyake, R., Yamazaki, M., 2012. An intelligent ship handling simulator with automatic collision avoidance function of target ships. Proc. of INSLC 17.
Huang, Y., Chen, L., Chen, P., Negenborn, R.R., Van Gelder, P.H.A.J.M., 2020. Ship collision avoidance methods: state-of-the-art. Saf. Sci. 121, 451–473.
Kim, D.J., Yun, K., Park, J.Y., Yeo, D.J., Kim, Y.G., 2019. Experimental investigation on turning characteristics of KVLCC2 tanker in regular waves. Ocean. Eng. 175, 197–206.
Kim, I.T., Kim, C., Kim, S.H., Ko, D., Moon, S.H., Park, H., Jin, B., 2021. Estimation of the manoeuvrability of the KVLCC2 in calm water using free running simulation based on CFD. Int. J. Nav. Archit. Ocean Eng. 13, 466–477.
Kim, D., Song, S., Jeong, B., Tezdogan, T., 2021a. Numerical evaluation of a ship's manoeuvrability and course keeping control under various wave conditions using CFD. Ocean Eng. 237, 109615.
Kim, D., Song, S., Jeong, B., Tezdogan, T., 2021b. Numerical evaluation of a ship's manoeuvrability and course keeping control under various wave conditions using CFD. Ocean Eng. 237, 109615.
Kim, D., Song, S., Jeong, B., Tezdogan, T., Incecik, A., 2021c. Unsteady RANS CFD simulations of ship manoeuvrability and course keeping control under various wave height conditions. Appl. Ocean Res. 117, 102940.
Kim, D., Song, S., Tezdogan, T., 2021d. Free running CFD simulations to investigate ship manoeuvrability in waves. Ocean Eng. 236, 109567.
Kim, D., Song, S., Tezdogan, T., 2021e. Free running CFD simulations to investigate ship manoeuvrability in waves. Ocean Eng. 236, 109567.
Kim, D., Tezdogan, T., 2022. CFD-based hydrodynamic analyses of ship course keeping control and turning performance in irregular waves. Ocean Eng. 248, 110808.
Kim, D., Yim, J., Song, S., Demirel, Y.K., Tezdogan, T., 2022a. A systematic investigation on the manoeuvring performance of a ship performing low-speed manoeuvres in adverse weather conditions using CFD. Ocean Eng. 263, 112364.
Kim, D., Song, S., Sant, T., Demirel, Y.K., Tezdogan, T., 2022b. Nonlinear URANS model for evaluating course keeping and turning capabilities of a vessel with propulsion system failure in waves. Int. J. Nav. Archit. Ocean Eng. 14, 100425.
Kim, D., Tezdogan, T., Incecik, A., 2022c. A high-fidelity CFD-based model for the prediction of ship manoeuvrability in currents. Ocean Eng. 256, 111492.
Kim, D., Tezdogan, T., Incecik, A., 2022d. Hydrodynamic analysis of ship manoeuvrability in shallow water using high-fidelity URANS computations. Appl. Ocean Res. 123, 103176.
Kim, D., Yim, J., Song, S., Park, J.B., Kim, J., Yu, Y., Tezdogan, T., 2023a. Path-following control problem for maritime autonomous surface ships (MASS) in adverse weather conditions at low speeds. Ocean Eng. 287, 115860.
Kim, D., Song, S., Turnock, S., Tezdogan, T., 2023b. Nonlinear URANS model for path-following control problem towards autonomous marine navigation under wave conditions. Ocean Eng. 270, 113681.
Kinaci, O.K., Ozturk, D., 2022. Straight-ahead self-propulsion and turning maneuvers of DTC container ship with direct CFD simulations. Ocean Eng. 244, 110381.
Lee, H.J., Furukawa, Y., 2022. Timing of stand-on ship actions for cooperative collision avoidance under calm water condition. Ocean Eng. 263, 112340.

Lee, H.J., Park, D.J., 2022. SASD modeling using an ANFIS to prevent the collision of MASS in restricted areas. J. Mar. Sci. Eng. 10 (7), 961.
Lee, H.J., Park, D.J., 2024. Collision evasive action timing for MASS using CNN–LSTM-based ship trajectory prediction in restricted area. Ocean Eng. 294, 116766.
Lee, H.J., Furukawa, Y., Park, D.J., 2021. Seafarers' awareness-based domain modelling in restricted areas. J. Navig. 74 (5), 1172–1188.
Musdscari, R., Broglia, R., Di Mascio, A., 2008. Trajectory prediction of a self-propelled hull by unsteady RANS computations. In: Proceedings of the 27th ONR Sym- Posium on Naval Hydrodynamics. Korea, Seoul.
Menter, F.R., 1994. Two-equation eddy-viscosity turbulence models for engineering applications. AIAA J. 32 (8), 1598–1605.
Mofidi, A., Carrica, P.M., 2014. Simulations of zigzag maneuvers for a container ship with direct moving rudder and propeller. Comput. Fluids 96, 191–203.
Namgung, Ho, 2022. Local route planning for collision avoidance of maritime autonomous surface ships in compliance with COLREGs rules sustainability, 14 (1), 198.
Porathe, T., 2019. Maritime Autonomous Surface Ships (MASS) and the COLREGS: do weneed quantified rules or is "the ordinary practice of seamen" specific enough? TransNav. International Journal on Marine Navigation and Safety of Sea Transportation 13 (3).
Sanada, Y., Tanimoto, K., Takagi, K., Gui, L., Toda, Y., Stern, F., 2013. Trajectories for ONR Tumblehome maneuvering in calm water and waves. Ocean. Eng. 72, 45–65.
Sanada, Y., Elshiekh, H., Toda, Y., Stern, F., 2019. ONR Tumblehome course keeping and maneuvering in calm water and waves. J.Mar. Sci. Technol. 24 (3), 948–967.
Shen, Z., Wan, D., Carrica, P.M., 2015. Dynamic overset grids in OpenFOAM with application to KCS self-propulsion and maneuvering. Ocean Eng. 108, 287–306.
Tam, C., Bucknall, R., 2010. Collision risk assessment for ships. J. Mar. Sci. Technol. 15, 257–270.
Wang, J., Zhao, W., Wan, D.C., 2016. Free maneuvering simulation of ONR Tumblehome using overset grid method in naoe-FOAM-SJTU solver. In: Proceedings of 31th Symposium on Naval Hydrodynamics.
Wang, B., He, Y., Hu, W., Mou, J., Li, L., Zhang, K., Huang, L., 2021. A decision-making method for autonomous collision avoidance for the stand-on vessel based on motion process and COLREGs. J. Mar. Sci. Eng. 9 (6), 584.
Wrobel, K., Gil, M., Huang, Y., Wawruch, R., 2022. The vagueness of COLREG versus collision avoidance techniques—a discussion on the current state and future challenges concerning the operation of autonomous ships. Sustainability 14 (24), 16516.
Yasukawa, H., Hasnan, M.A.A., Matsuda, A., 2021. Validation of 6-DOF motion simulations for ship turning in regular waves. J. Mar. Sci. Technol. 1–16.
Zhou, X.Y., Huang, J.J., Wang, F.W., Wu, Z.L., Liu, Z.J., 2020. A study of the application barriers to the use of autonomous ships posed by the good seamanship requirement of COLREGs. J. Navig. 73 (3), 710–725.